A LOOK AT CHEBYSHEV'S BIAS AND THE PRIME RACE

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Abstract. Although the Prime Number Theorem for arithmetic progressions guarantees the asymptotic equidistribution of primes among coprime residue classes, computational observations reveal a persistent asymmetry known as Chebyshev's bias. In this paper, I utilized numerical simulations to examine this phenomenon across several moduli. I further analyze these findings through the theoretical framework of Rubinstein and Sarnak, discussing how the Generalized Riemann Hypothesis and the distribution of zeros of Dirichlet L-functions account for this "rigged" prime race. While primes are balanced in the infinite limit, there is preference for non-residue classes in finite ranges.

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1. Introduction

In 1853, Pafnuty Chebyshev sent a letter to Nikolaj Fuss in which he observed that primes congruent to 3 (mod 4) seem to outnumber those congruent to 1 (mod 4) [1]. This phenomenon is now known as Chebyshev's bias. Let $\pi(x; q, a)$

denote the number of primes $p \leq x$ with $p \equiv a \pmod{q}$. I study the behavior of $\psi(x; q, a, b) = \pi(x; q, a) - \pi(x; q, b)$ across several moduli, beginning with two-lane races and generalizing to races with more lanes.

2. Preliminaries

Definition 2.1 (Prime Counting Function)**.** The *prime counting function* $\pi(x)$ denotes the number of primes $p \leq x$. More generally, $\pi(x; q, a)$ denotes the number of primes $p \leq x$ such that $p \equiv a \pmod{q}$.

Theorem 2.2 (Dirichlet's Theorem on Primes in Arithmetic Progressions)**.** *Let a and q be positive integers with* $\gcd(a, q) = 1$*. Then there are infinitely many primes p such that $p \equiv a \pmod{q}$.*

Theorem 2.3 (Prime Number Theorem for Arithmetic Progressions)**.** *Let a and q be integers such that* $\gcd(a, q) = 1$*. As $x \to \infty$:*

$$\pi(x, q, a) \sim \frac{Li(x)}{\phi(q)} \sim \frac{1}{\phi(q)} \frac{x}{\log x}$$

where $\phi(q)$ is the Euler totient function and $Li(x) = \int_2^x \frac{dt}{\log t}$.

Remark 2.4. Dirichlet's theorem ensures that every valid lane in the race (those where $\gcd(a, q) = 1$) contains infinitely many primes, meaning no lane ever vanishes. The Prime Number Theorem goes a step further by confirming that eventually, every lane receives an equal share of the contestants. The interesting question, therefore, is not whether each lane contains infinitely many primes, but whether certain residue classes maintain a consistent lead over others across finite initial segments.

Definition 2.5 (Prime Race Function)**.** For a modulus q and residues a, b coprime to q, the *prime race function* is

$$\psi(x; q, a, b) = \pi(x; q, a) - \pi(x; q, b).$$

When $\psi > 0$, the class $a \pmod{q}$ is leading. When $\psi < 0$, the class $b \pmod{q}$ is leading.

3. Simulations

I ran computational simulations of $\psi(x; q, a, b)$ for six moduli, up to a limit of $x = 10^7$. For each modulus q, I plotted every pairwise race between residue classes coprime to q. Pairs where one class is a quadratic residue and the other is not are drawn at full opacity, as these are the races where Chebyshev's bias is predicted to be most visible. All other pairs are shown at reduced opacity for reference. The code implementation is given in Appendix 5.1.

3.1. **Two-Lane Races.** The moduli $q = 3, 4,$ and 6 are special since $\phi(q) = 2$ for each, meaning there are only two lanes and one race to observe.

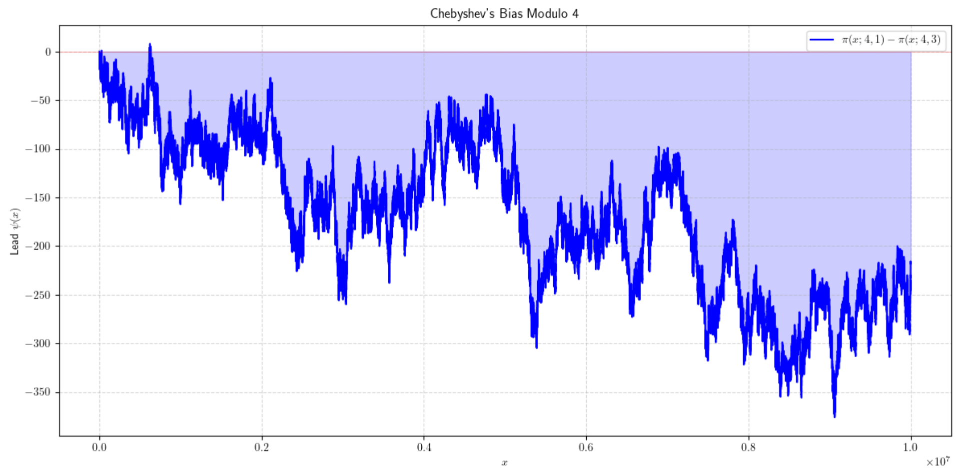

FIGURE 1. The prime race modulo 4, between residue classes 1 (mod 4) and 3 (mod 4).

In the modulo 4 race (Figure 1), the 3 (mod 4) class leads for nearly the entire range. The function $\psi(x; 4, 1, 3)$ stays predominantly negative, peeking above zero only briefly near the start.

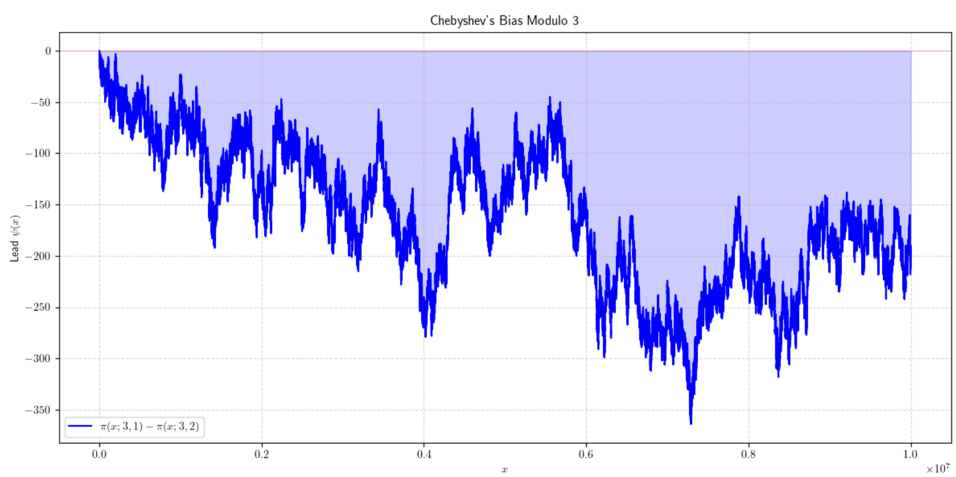

FIGURE 2. The prime race modulo 3, between residue classes 1 (mod 3) and 2 (mod 3).

In the modulo 3 race (Figure 2), the 2 (mod 3) class similarly dominates throughout. The lead grows over time, suggesting a persistent and widening bias.

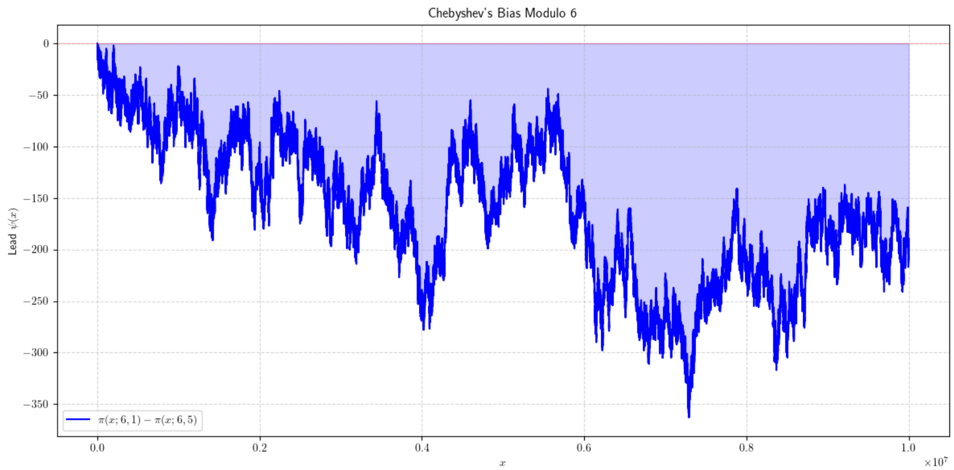

FIGURE 3. The prime race modulo 6, between residue classes $1 \pmod 6$ and $5 \pmod 6$.

The modulo 6 race (Figure 3) shows the same pattern, where the class $5 \pmod 6$ leads throughout. Together, I see that the three two-lane races all exhibit the same bias.

3.2. **Several-Lane Races.** Now I examine moduli with more than two coprime residue classes, where multiple races occur simultaneously.

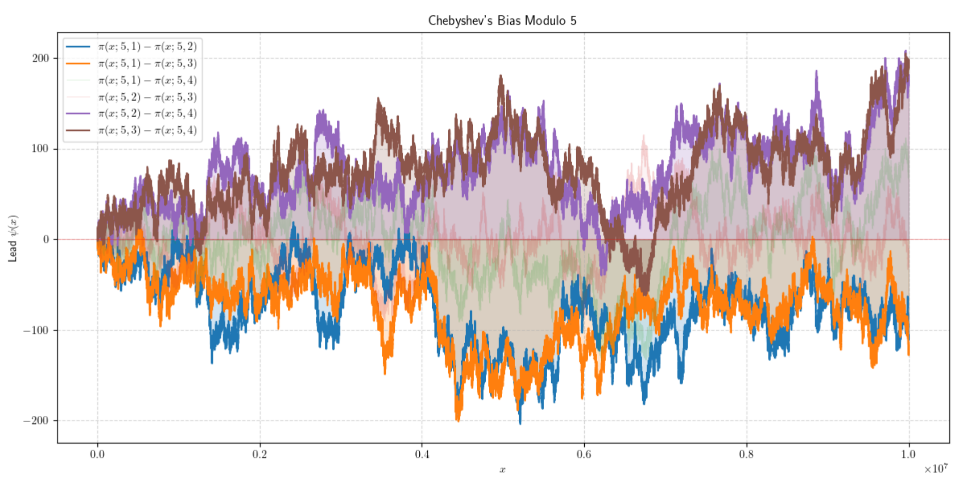

FIGURE 4. The prime race modulo 5, among residue classes $1, 2, 3, 4 \pmod 5$.

For $q = 5$ (Figure 4), the quadratic residues modulo 5 are 1 and 4 (since $1^2 \equiv 1$ and $2^2 \equiv 4$), while 2 and 3 are non-residues. The highlighted races, those between a residue and a non-residue, show a consistent bias in favor of the non-residue class.

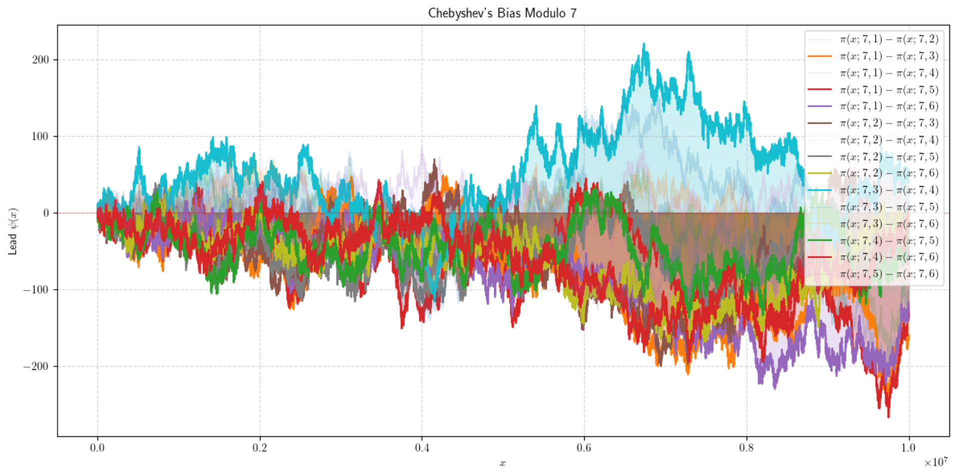

FIGURE 5. The prime race modulo 7, among residue classes $1, 2, 3, 4, 5, 6 \pmod 7$.

For $q = 7$ (Figure 5), the quadratic residues are $1, 2, 4$ (since $1^2 \equiv 1$, $3^2 \equiv 2$, $2^2 \equiv 4 \pmod 7$), and the non-residues are $3, 5, 6$. Again, the highlighted pairs show the tendency for non-residue classes to dominate.

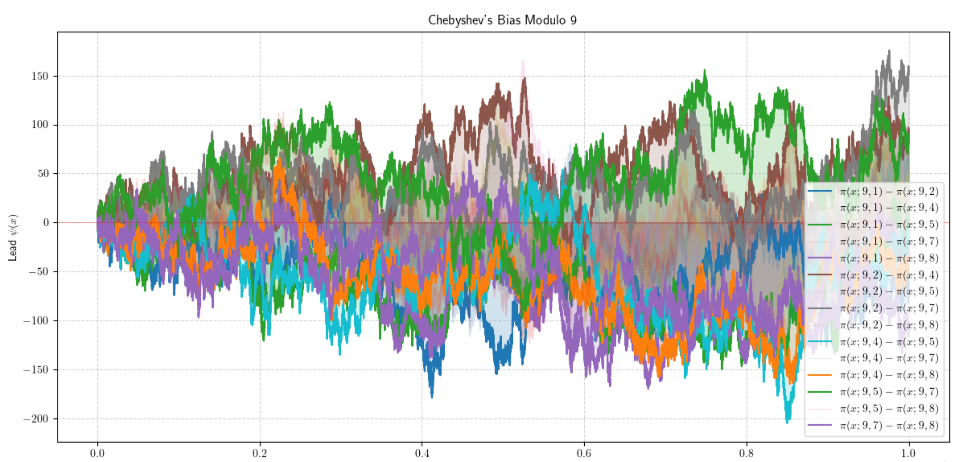

FIGURE 6. The prime race modulo 9, among coprime residue classes modulo 9.

For $q = 9$ (Figure 6), the coprime residues are $1, 2, 4, 5, 7, 8$, of which $1, 4, 7$ are quadratic residues. The pattern persists where non-residue classes tend to lead their respective races.

4. Analysis

Across all six simulations, a clear pattern occurs. In every race between a quadratic residue and a quadratic non-residue, the non-residue class tends to win. This is precisely the phenomenon Chebyshev first noticed, and it is not a coincidence.

4.1. Why the Bias Exists.

The key to understanding the bias lies in the structure of the error term in the prime counting function. By the Prime Number Theorem for arithmetic progressions, the counts $\pi(x; q, a)$ for different residues a are asymptotically equal, since each class receives roughly $\frac{1}{\phi(q)}$ of all primes. So in the long run, all lanes are equally matched.

However, Rubinstein and Sarnak showed that the fluctuations around this equal distribution are not symmetric [2]. Under the Generalized Riemann Hypothesis (GRH), the normalized race function has a limiting logarithmic distribution $\mu_{q;a,b}$, and this distribution has a nonzero mean. The mean is determined by a bias constant

$$c(q, a) = -1 + \#\{x \pmod q : x^2 \equiv a \pmod q\},$$

which counts how many square roots a has modulo q, minus one.

4.2. Squares Versus Non-Squares.

For a quadratic residue a, there exist x such that $x^2 \equiv a \pmod q$, so $c(q, a) \geq 0$. For a quadratic non-residue b, no such x exists, so $c(q, b) = -1$. This asymmetry shifts the mean of the limiting distribution: non-residue classes have a negative bias constant, which corresponds to a *positive* mean in the race function $\psi(x; q, b, a)$. This is exactly why non-residues tend to lead.

This is verified this in each of my cases. For $q = 4$: the residue 1 satisfies $1^2 \equiv 1 \pmod 4$, so $c(4, 1) = 0$, while 3 is a non-residue so $c(4, 3) = -1$. The distribution for $\psi(x; 4, 1, 3)$ is thus centered at a positive mean, consistent with Figure 1. The same reasoning applies to $q = 3$ and $q = 6$, where 1 is always a square and $2 \pmod 3$, $5 \pmod 6$ are non-squares, explaining the bias seen in Figures 2 and 3.

4.3. Is the Bias Permanent?

No. In 1914, Littlewood proved that $\psi(x; 4, 1, 3)$ changes sign infinitely often [3], meaning the $1 \pmod 4$ class does eventually take the lead. However, this happens very rarely within normal computational range. This result extends to all such races. What Chebyshev's bias describes is not a permanent advantage but a probabilistic one. If an integer x is picked at random, it is far more likely that $\pi(x; q, \text{non-residue}) > \pi(x; q, \text{residue})$.

Rubinstein and Sarnak made this precise by computing the proportion of the time each class leads. Formally, for $\psi(x; q, a, b) = \pi(x; q, a) - \pi(x; q, b)$, define the logarithmic density

$$\delta(q; a, b) = \lim_{X \to \infty} \frac{1}{\log X} \int_2^X \mathbf{1}[\pi(x; q, a) > \pi(x; q, b)] \, \frac{dx}{x},$$

which equals $\delta(P_{q;a,b})$ in the notation of [2], where $P_{q;a,b} = \{x \geq 2 : \pi(x; q, a) > \pi(x; q, b)\}$. For the modulo 4 race, they computed $\delta(4; 3, 1) \approx 0.9959$ [2], meaning the 3 (mod 4) class leads about 99.59% of the time. This is consistent with what my simulation shows.

5. Conclusion

In this paper, I explored the phenomenon of Chebyshev's bias through computational simulation and theoretical discussion. I observed across six moduli; $q = 3, 4, 5, 6, 7, 9$; that primes congruent to quadratic non-residues consistently tend to lead those congruent to quadratic residues over the range $[2, 10^7]$.

The Prime Number Theorem for Arithmetic Progressions (Theorem 2.3) tells us that in the long run, all residue classes receive an equal share of the primes. In this sense, the race is fair. Yet the simulations reveal that fairness in the limit does not mean fairness on any finite scale. The bias constant $c(q, a)$, determined by the number of square roots of a modulo q, systematically disadvantages quadratic residues by shifting the mean of the limiting distribution. Littlewood's theorem reminds us that the lead does change infinitely often, but Rubinstein and Sarnak quantify just how rare those reversals are. For the modulo 4 race, the 3 (mod 4) class leads approximately 99.59% of the time.

To me, it is interesting how it took humans over a century of deep mathematics to only partially explain this seemingly trivial phenomenon that was observed all the way back in 1853. Until the Riemann Hypothesis is proven, the proof of this phenomenon is still as fragile as it can be.

Appendix

5.1. **Code Implementation.**

```
import math
import sympy
import matplotlib.pyplot as plt
import warnings
plt.rcParams['text.usetex'] = True
plt.rcParams['text.latex.preamble'] = r'\usepackage{amsmath}'
warnings.filterwarnings("ignore")

def possible_mod(q):
    '''Return a list of all residues coprime to q'''
    return [x for x in range(1, q) if math.gcd(x, q) == 1]

def is_square(a, q):
```

```
    '''Return True if a is a square mod q'''
    return any((x * x) % q == a % q for x in range(1, q))

def is_qr_pair(a, b, q):
    '''Return True if this pair is a (non-square, square) race'''
    a_is_qr = is_square(a, q)
    b_is_qr = is_square(b, q)
    return a_is_qr != b_is_qr

def plot_psi(q, limit):
    '''Plot the races.'''
    lanes = possible_mod(q)
    primes = list(sympy.primerange(2, limit + 1))

    history = {a: [0] for a in lanes}
    x_coords = [0]
    current_counts = {a: 0 for a in lanes}

    for p in primes:
        res = p % q
        if res in current_counts:
            current_counts[res] += 1
        x_coords.append(p)
        for a in lanes:
            history[a].append(current_counts[a])

    pairs = [(lanes[i], lanes[j]) for i in range(len(lanes))
                for j in range(i+1, len(lanes))]
    single = len(pairs) == 1
    colors = plt.cm.tab10.colors

    fig, ax = plt.subplots(figsize=(12, 6))

    for idx, (a, b) in enumerate(pairs):
        psi = [h_a - h_b for h_a, h_b in
                zip(history[a], history[b])]
        color = 'blue' if single else colors[idx % len(colors)]
        key_pair = single or is_qr_pair(a, b, q)
```

```
        ax.step(x_coords, psi, where='post', color=color,
                alpha=1 if key_pair else 0.15,
                linewidth=1.5 if key_pair else 0.7,
                label=rf'$\pi(x;{q},{a}) - \pi(x;{q},{b})$')
        ax.fill_between(x_coords, psi, step='post',
                        alpha=0.2 if key_pair else 0.05,
                        color=color)

    ax.axhline(0, color='red', linestyle='-',
               linewidth=0.8, alpha=0.3)
    ax.set_xlabel(r'$x$')
    ax.set_ylabel(r'Lead $\psi(x)$')
    ax.set_title(f"Chebyshev's Bias Modulo {q}")
    ax.legend()
    ax.grid(True, linestyle='--', alpha=0.5)
    plt.tight_layout()
    plt.show()
```

5.2. **Proof of Dirichlet's Theorem.**

Theorem 5.1 (Dirichlet's Theorem on Primes in Arithmetic Progressions)**.** *Let a and q be positive integers with* $\gcd(a, q) = 1$*. Then there are infinitely many primes p such that $p \equiv a \pmod q$.*

Proof. An elementary proof is given in Wang [4]. □

5.3. **Proof of Prime Number Theorem for Arithmetic Progressions.**

Theorem 5.2 (Prime Number Theorem for Arithmetic Progressions)**.** *Let a and q be integers such that* $\gcd(a, q) = 1$*. As $x \to \infty$:*

$$\pi(x, q, a) \sim \frac{Li(x)}{\phi(q)} \sim \frac{1}{\phi(q)} \frac{x}{\log x}$$

where $\phi(q)$ is the Euler totient function and $Li(x) = \int_2^x \frac{dt}{\log t}$.

Proof. We follow the elementary approach of Selberg [5], adapted to arithmetic progressions. The starting point is Selberg's asymptotic formula

$$\sum_{p \le x} \log^2 p + \sum_{pq \le x} \log p \log q = 2x \log x + O(x),$$

which Selberg notes can be extended to arithmetic progressions using the orthogonality of Dirichlet characters. Specifically, one can establish an analogous formula for each residue class $a \pmod q$ with $\gcd(a, q) = 1$, where the main term picks up a factor of $1/\phi(q)$ from the character decomposition $\mathbf{1}_{n \equiv a} = \frac{1}{\phi(q)} \sum_{\chi \pmod q} \bar{\chi}(a)\chi(n)$.

Writing $\vartheta(x;q,a) = \sum_{\substack{p\le x\\ p\equiv a}} \log p = \frac{x}{\phi(q)} + R(x;q,a)$, the error term R satisfies the same self-referential contraction inequality as in the classical case, and the same iterative argument forces $R(x;q,a) = o(x)$. Hence $\vartheta(x;q,a) \sim x/\phi(q)$, and converting to $\pi(x;q,a)$ via partial summation gives $\pi(x;q,a) \sim \mathrm{Li}(x)/\phi(q) \sim x/(\phi(q)\log x)$. $\square$

References

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[3] J. E. Littlewood, *Sur la distribution des nombres premiers*, Comptes Rendus, 158 (1914), 1869–1872.

[4] Z. Wang, *Elementary Proof of Dirichlet's Theorem*, University of Chicago REU, 2017. [PDF]

[5] A. Selberg, *An elementary proof of the prime-number theorem*, Ann. of Math. (2) 50 (1949), no. 2, 305–313.